

Youtube

Functional requirements:

user should be able to watch any videos based on his preference,
user should be able to search any videos among the existing videos based on preference,
Users feed should be updated as per activity,
videos resolution should get adjusted based on the network bandwidth

Non-functional requirements:

data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput ,
CAP theorem(here eventual consistency is fine , but here we need to prioritise 99.99% uptime.
I.e Availability>>Consistency)

Core Entities:

videos,
users,
history,
feed

API :

POST:/upload

```
{  
  userID,  
  filemetadata  
}
```

GET:/search/{videoID}

HLD :

Presigned URL

